Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -230.75351694
E1 (Ha) : -230.57821527
Gap (Ha) : 0.17530166
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : -0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961507
<D> / site : 0.592301

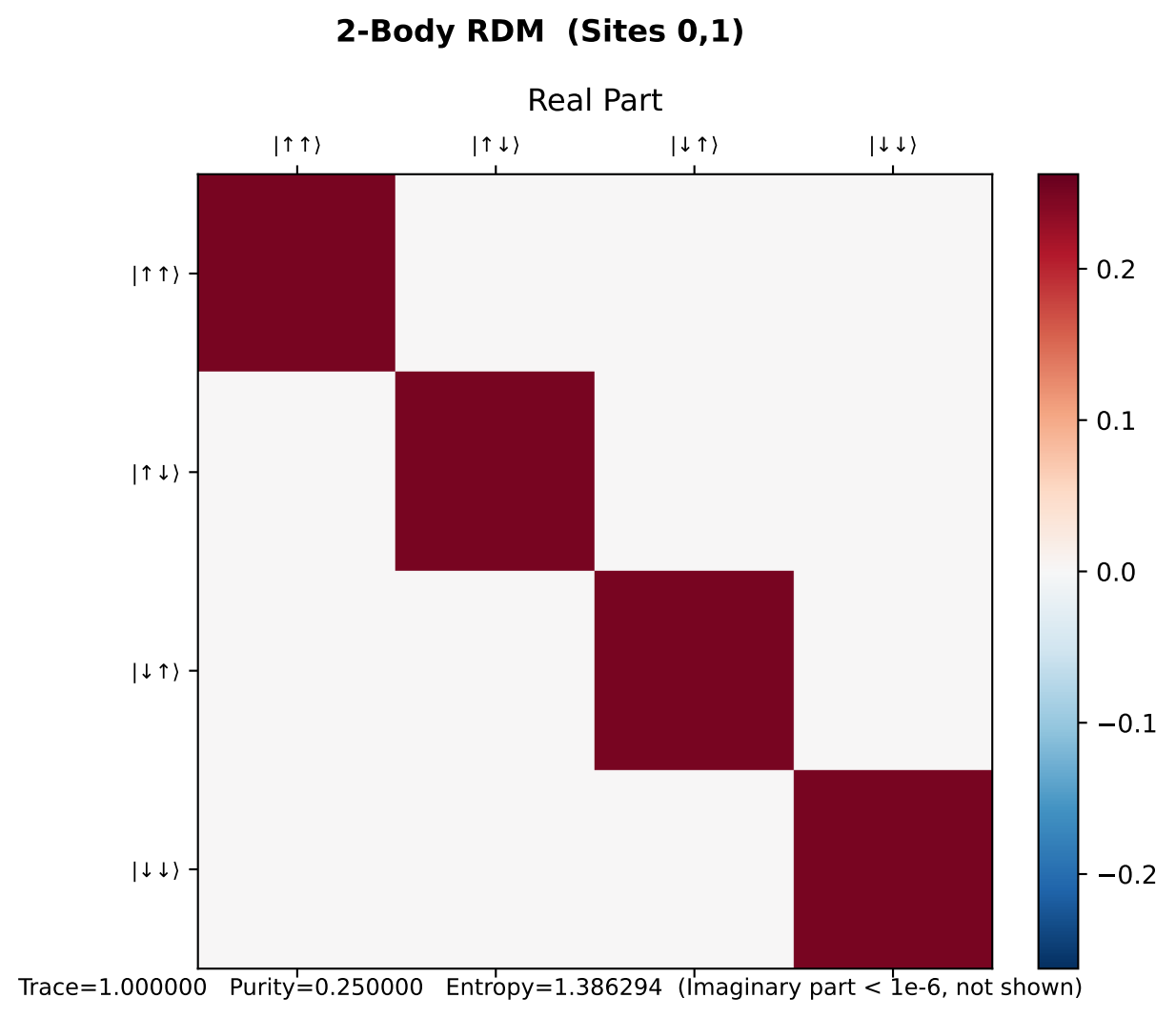
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Body order : 2
Clusters   : 10

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Summary — 2-Body RDMs (Benzene, CAS(6,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1	0.250000	—	—	—	—	—	MIXED
0,2	0.250000	—	—	—	—	—	MIXED
0,3	0.250000	—	—	—	—	—	MIXED
0,4	0.250000	—	—	—	—	—	MIXED
1,2	0.250000	—	—	—	—	—	MIXED
1,3	0.297715	—	—	—	—	—	MIXED
1,4	0.251106	—	—	—	—	—	MIXED
2,3	0.251113	—	—	—	—	—	MIXED
2,4	0.297588	—	—	—	—	—	MIXED
3,4	0.251897	—	—	—	—	—	MIXED


2-Body RDM (Sites 0,1)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

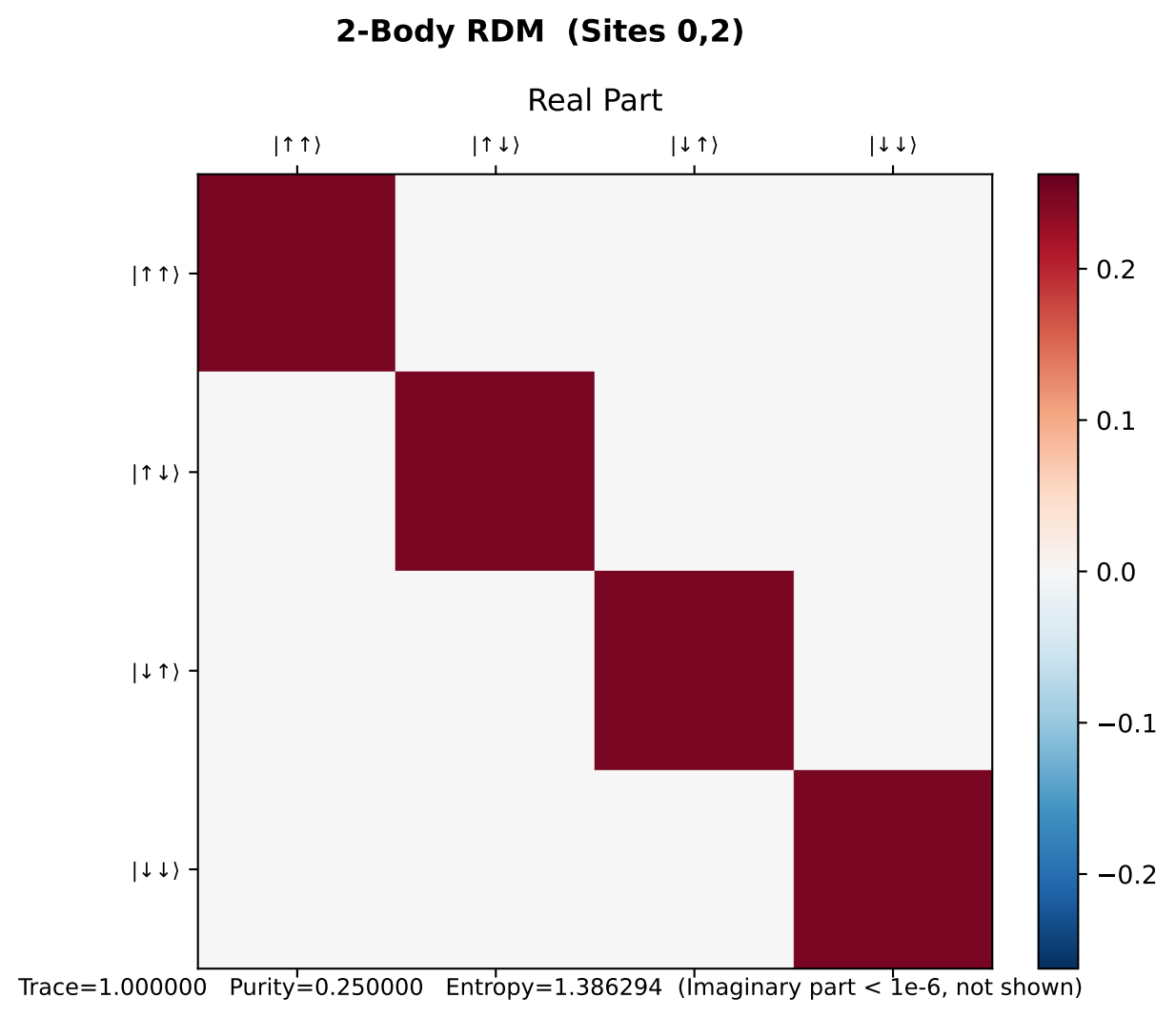
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

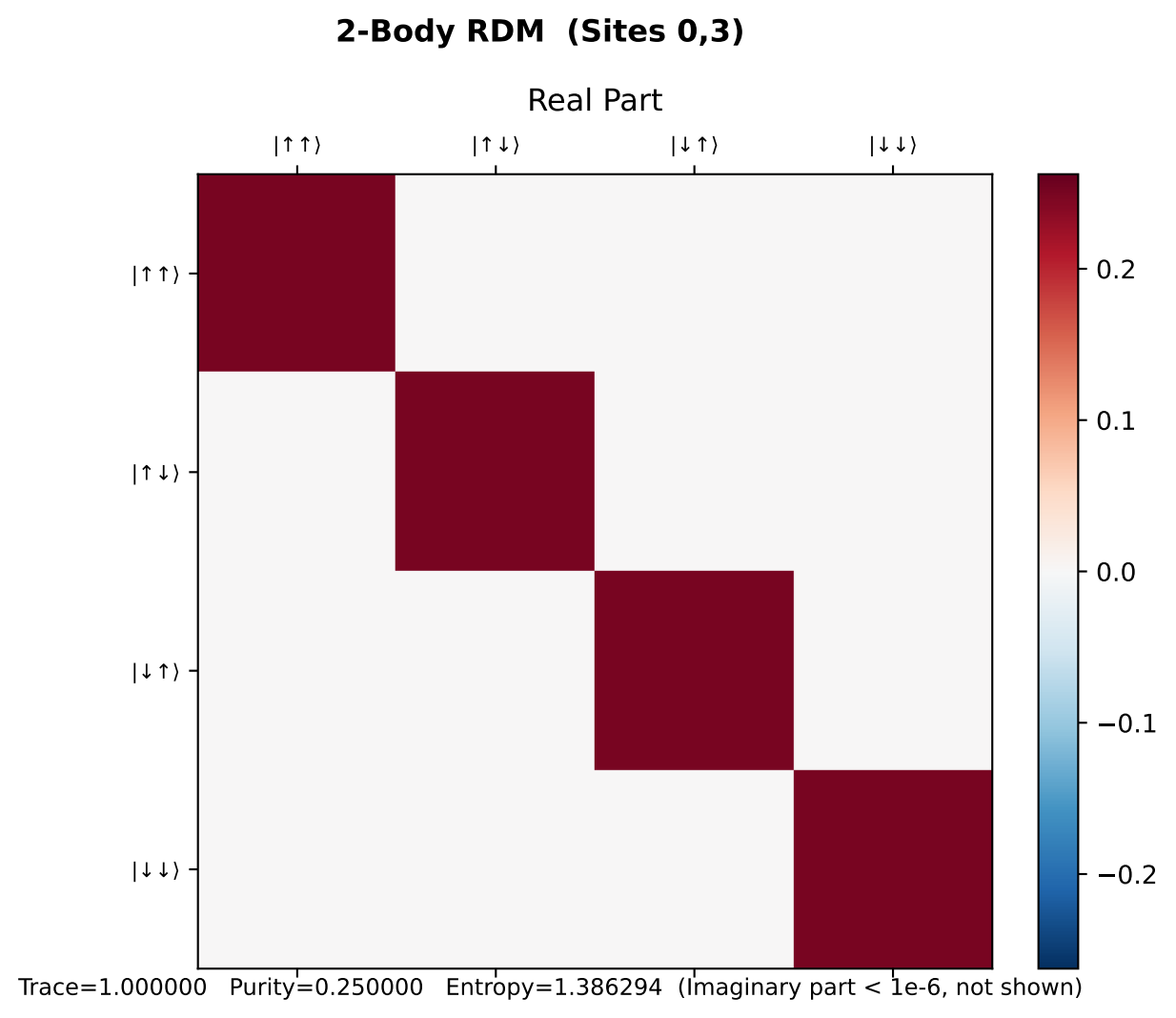
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

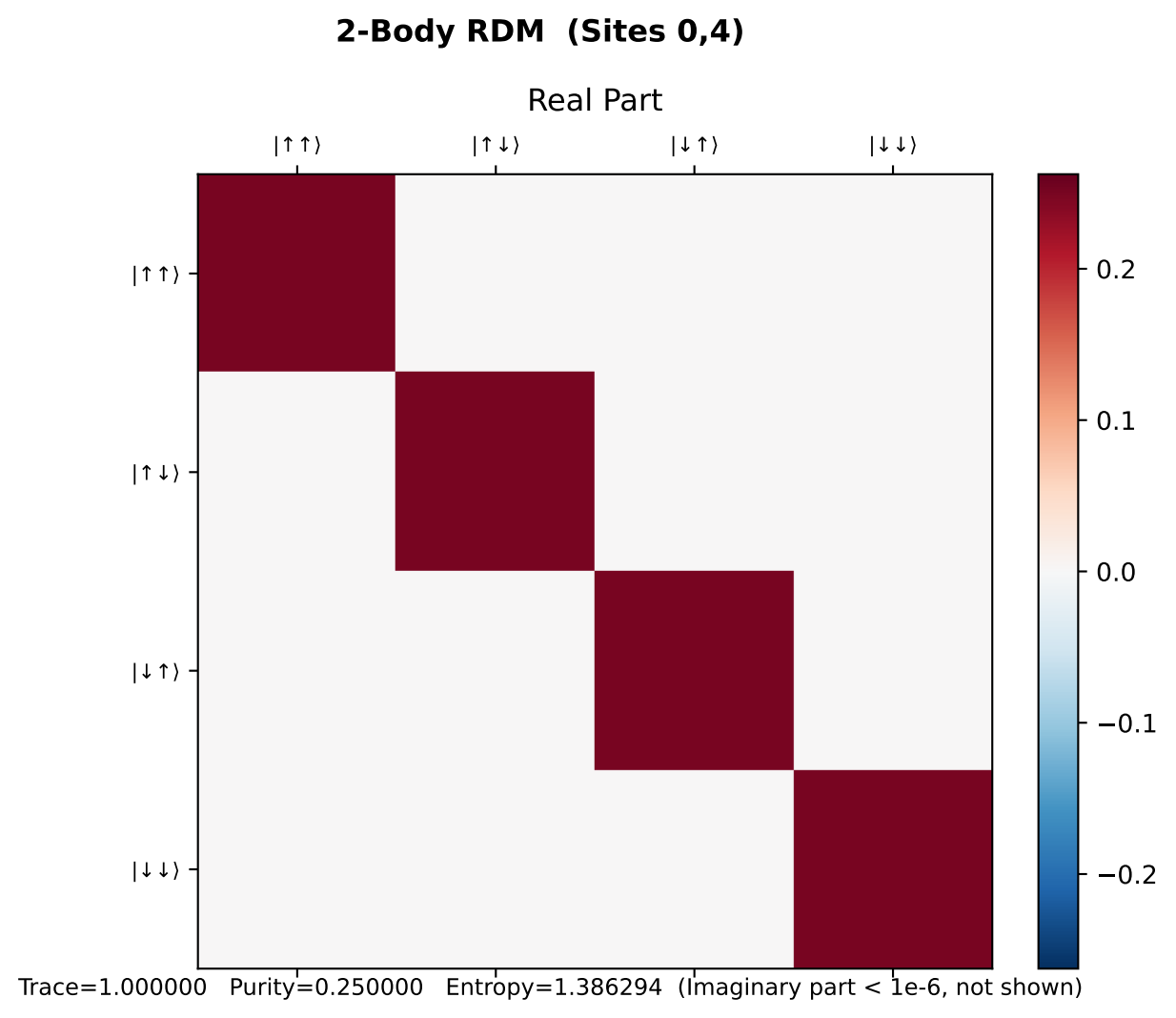
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

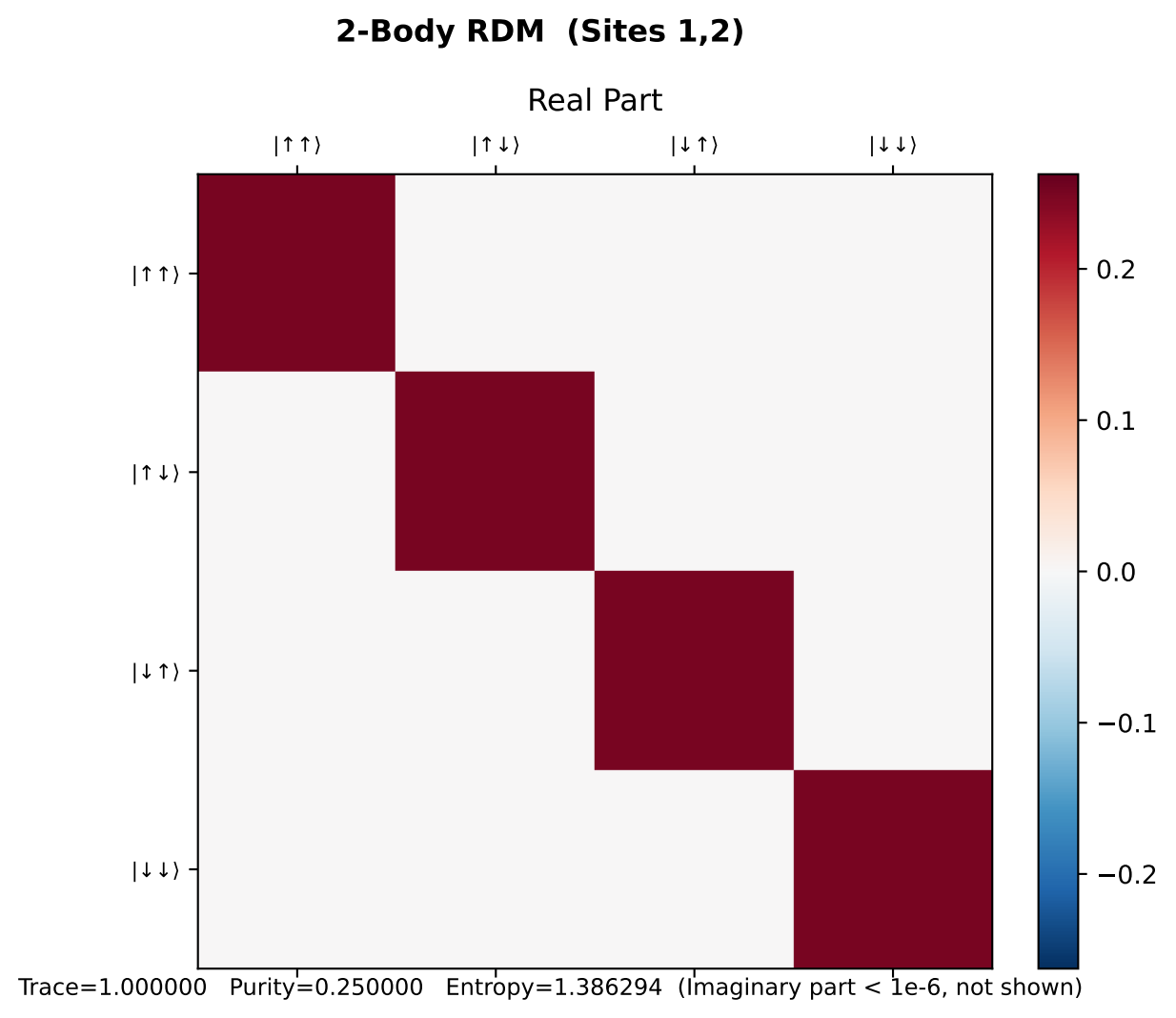
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000696
Von Neumann entropy	: 1.3862942219
Eigenvalue min	: 2.497716e-01
Eigenvalue max	: 2.500761e-01
Imaginary part	: < 1e-6 (not shown)

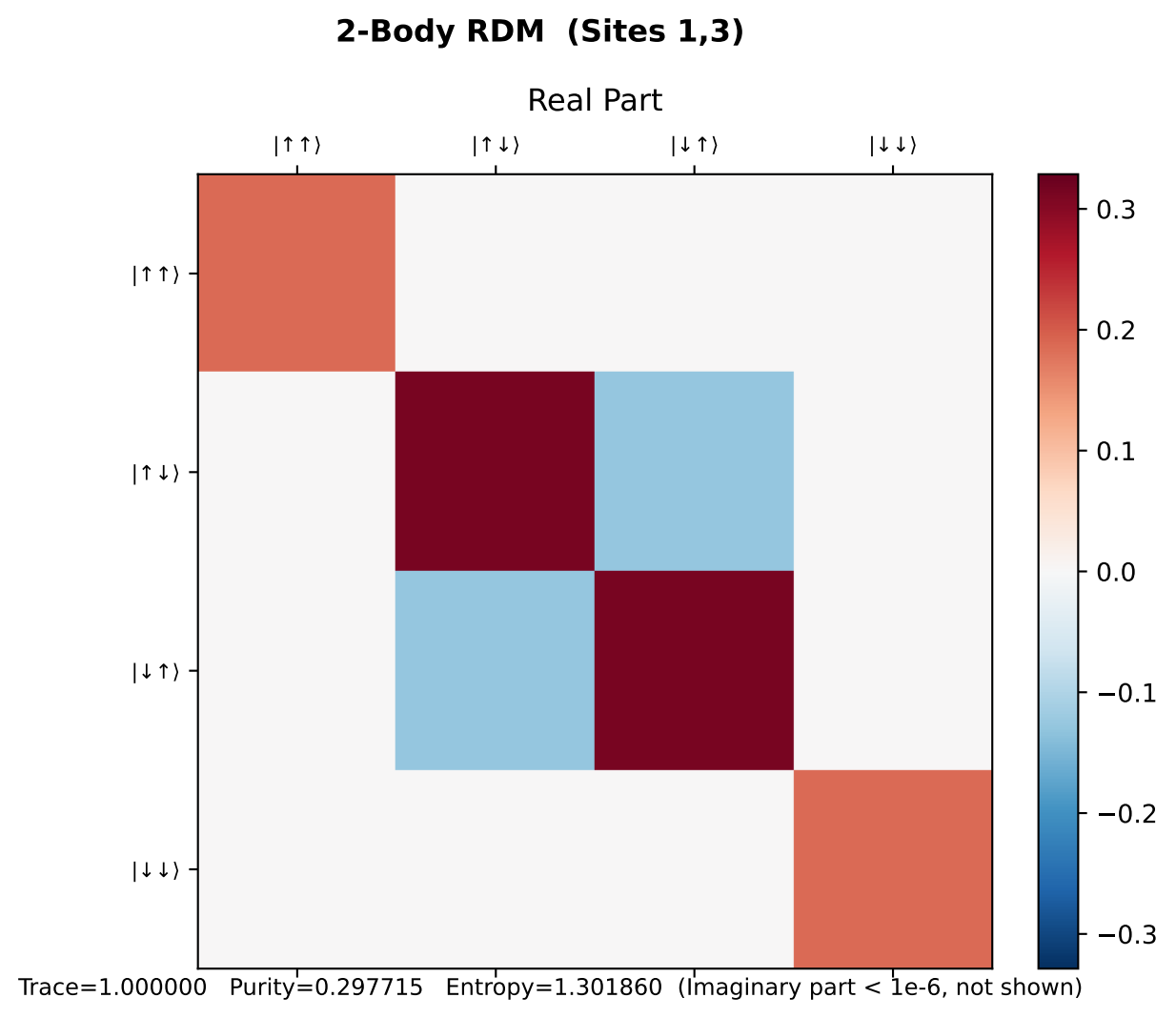
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000007

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.297715 Entropy=1.301860 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2977153042
Von Neumann entropy	: 1.3018602828
Eigenvalue min	: 1.869423e-01
Eigenvalue max	: 4.391731e-01
Imaginary part	: < 1e-6 (not shown)

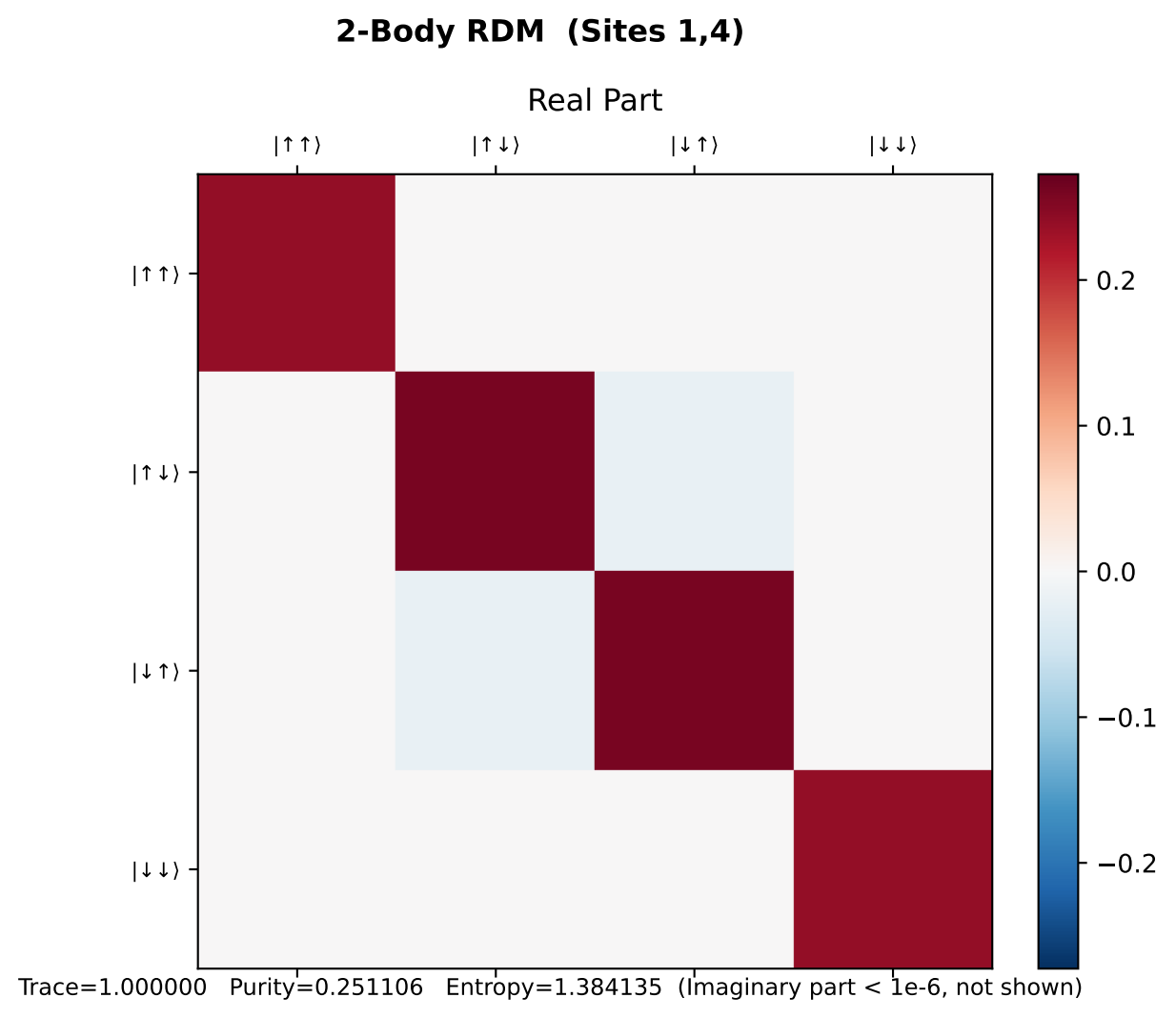
GME Analysis
==========

gamma = Tr(rho^2) = 0.29771530

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251106 Entropy=1.384135 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2511060653
Von Neumann entropy	: 1.3841352911
Eigenvalue min	: 2.403994e-01
Eigenvalue max	: 2.788019e-01
Imaginary part	: < 1e-6 (not shown)

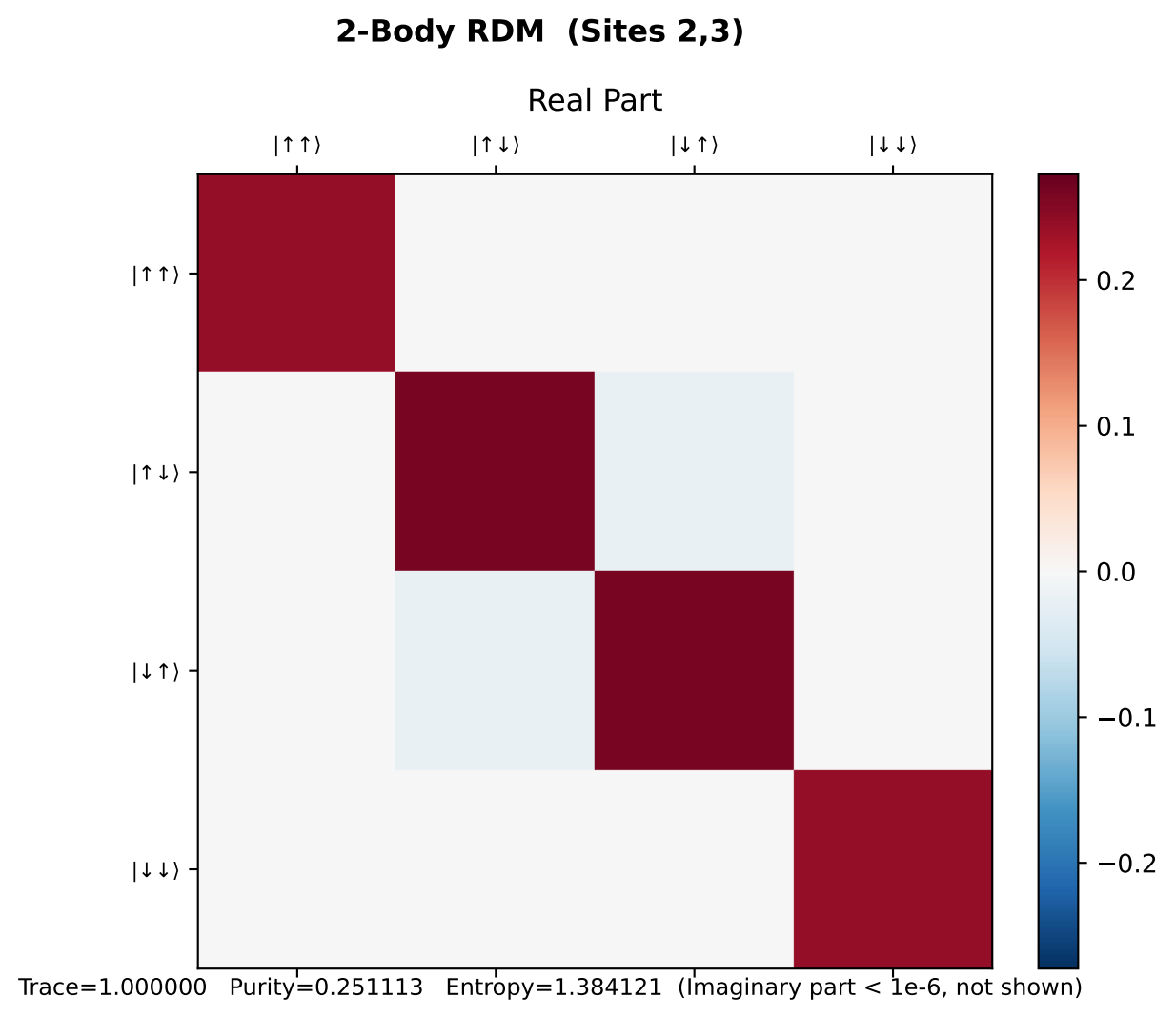
GME Analysis
==========

gamma = Tr(rho^2) = 0.25110607

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251113 Entropy=1.384121 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2511134215
Von Neumann entropy	: 1.3841210978
Eigenvalue min	: 2.403675e-01
Eigenvalue max	: 2.788975e-01
Imaginary part	: < 1e-6 (not shown)

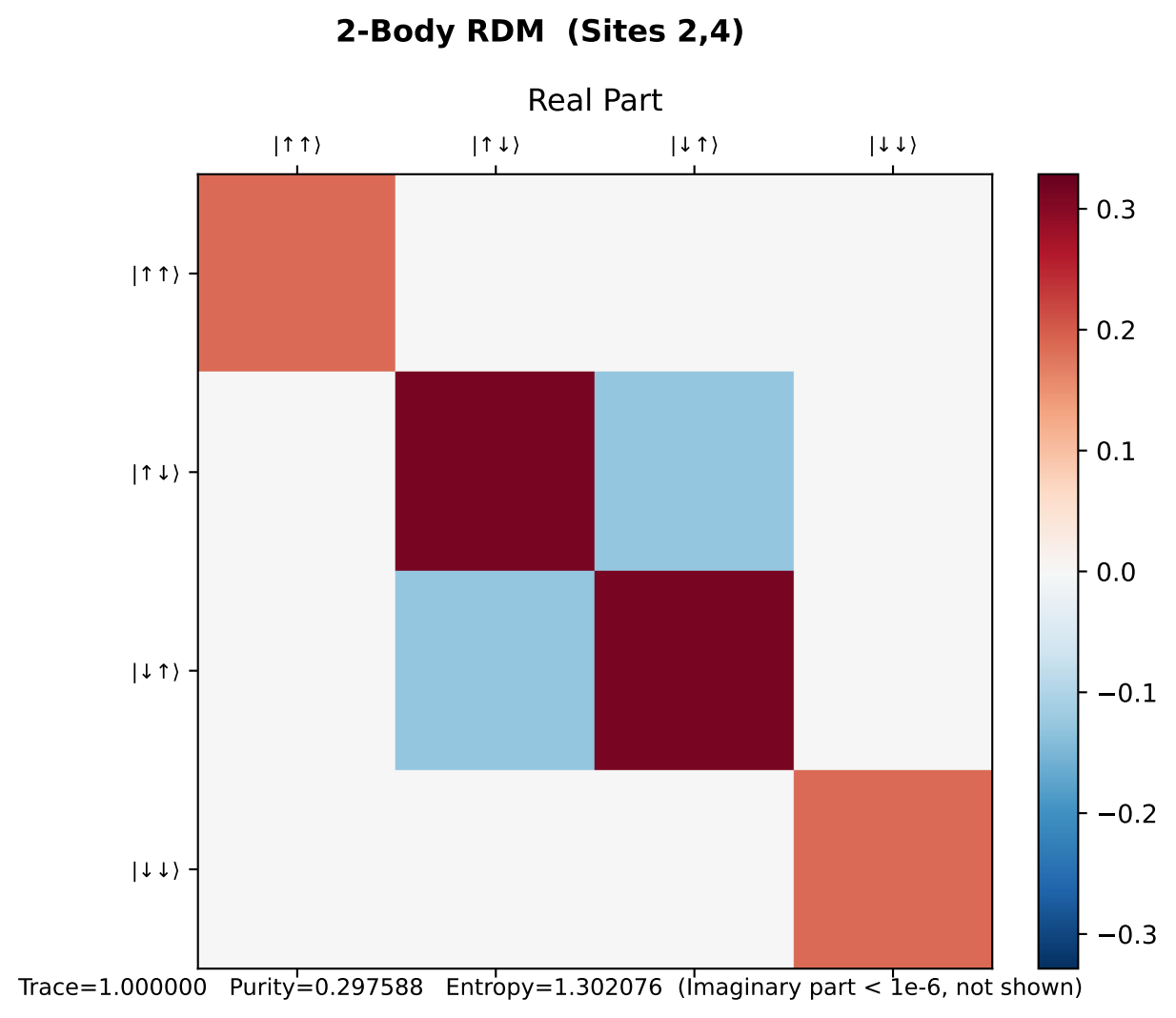
GME Analysis
==========

gamma = Tr(rho^2) = 0.25111342

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.297588 Entropy=1.302076 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2975876704
Von Neumann entropy	: 1.3020763916
Eigenvalue min	: 1.870267e-01
Eigenvalue max	: 4.389200e-01
Imaginary part	: < 1e-6 (not shown)

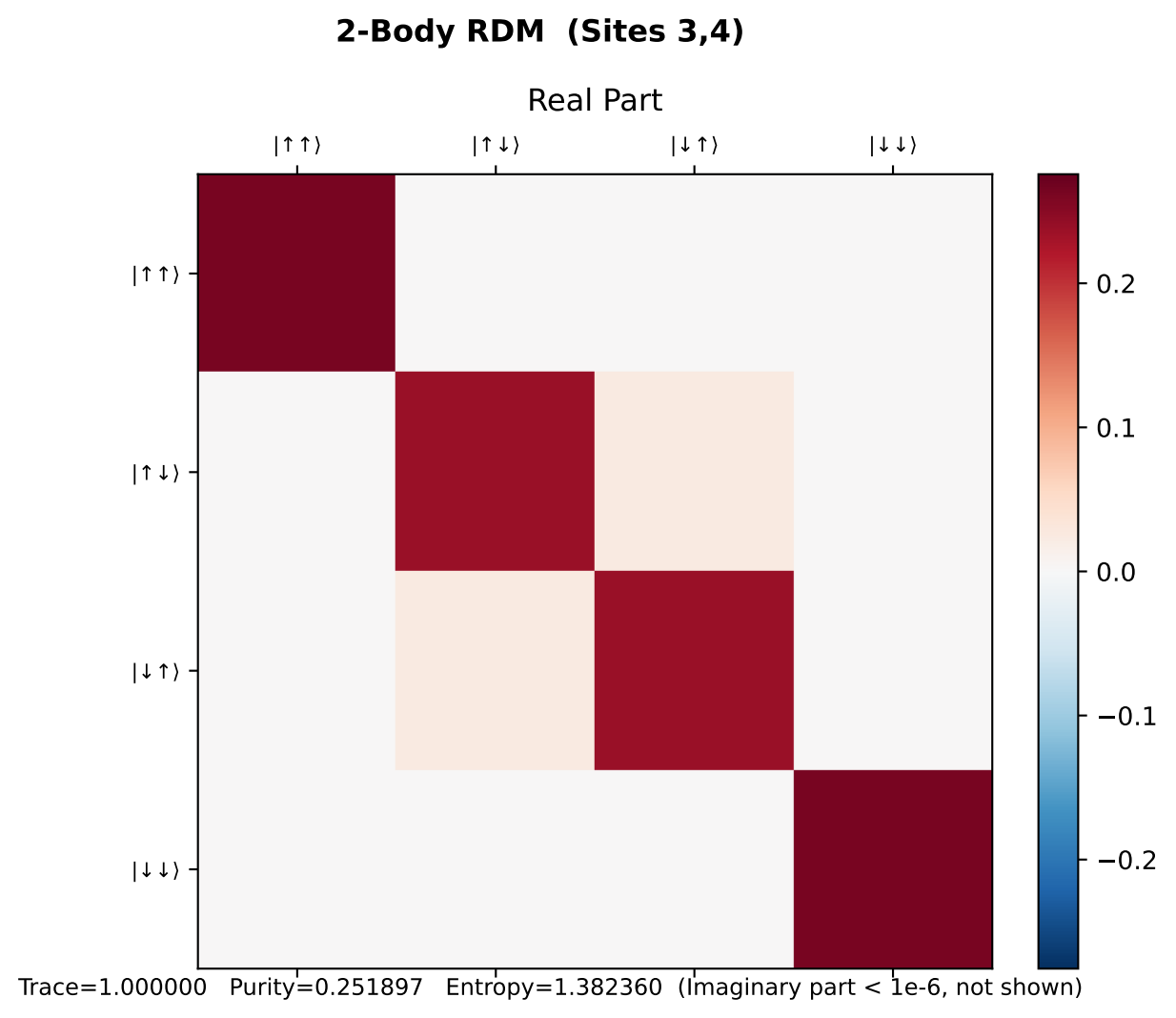
GME Analysis
==========

gamma = Tr(rho^2) = 0.29758767

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251897 Entropy=1.382360 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2518973524
Von Neumann entropy	: 1.3823601381
Eigenvalue min	: 2.122771e-01
Eigenvalue max	: 2.625743e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.25189735

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.